

Time Complexity (Big-O Notation)

```
// Example 1:
int a = 10;
int b = 20;
int sum = a + b;
```

Time complexity: $T(n) = O(1)$

```
// Example 2:
if (n % 2 == 0)
{
    cout << "Even";
}
else {
    cout << "Odd";
}
```

Time complexity: $T(n) = O(1)$

```
// Example: 3:
int sum = 0;
for (int i = 0; i < n; i++) {
    sum += arr[i];
}
```

Time complexity: $T(n) = O(n)$

```
// Example: 4:

int count = 0;
for (int i = 0; i < n; i++) {
    if (arr[i] % 2 == 0) {
        count++;
    }
}
```

Time complexity: $T(n) = O(n)$

```
// Example: 5:

bool found = false;
for (int i = 0; i < n; i++) {
```

```

        if (arr[i] == target) {
            found = true;
            break;
        }
    }
}

```

Time complexity: $T(n) = O(n)$

//Example: 6

```

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        printf("%d %d", i, j);
    }
}

```

Time complexity: $T(n) = O(n^2)$

//Example: 7

```

for (int i = 0; i < n; i++) {
    for (int j = i + 1; j < n; j++) {
        printf("%d %d", i, j);
    }
}

```

Time complexity: $T(n) = O(n^2)$

//Example: 8

```

int cnt = 0;
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        cnt++;
    }
}

```

Time complexity: $T(n) = O(n^2)$

// Example: 9

```

for (int i = 0; i < n; i++) {
    if (arr[i] > 100) {
        count++;
    }
}

```

Time complexity: $T(n) = O(n)$

```
// Example: 10
```

```
bool found = false;
for (int i = 0; i < n; i++) {
    if (arr[i] == target) {
        found = true;
        break;
    }
}
```

Time complexity: $T(n) = O(n)$
